

GalaxyAssistant: An Intelligent Assistant for Multimedia Event Retrieval

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Abstract. The exponential growth of large-scale multimedia data necessitates efficient event retrieval systems, a challenge addressed by competitions like LSC, VBS, and the Ho Chi Minh City AI Challenge. To address this, we, GALAXY-AI team, propose GalaxyAssistant, an intelligent assistant framework designed for in-depth analysis and information retrieval from complex video data. Our system’s intelligence is rooted in its tri-modal indexing, which concurrently processes video via three parallel pipelines following shot detection (TransNetV2). First, a visual-language model (SigLIP) generates dense visual embeddings. Second, an image captioning model (InternVL) creates textual descriptions. Third, an ASR model (ChunkFormer) transcribes the audio. Both caption and audio transcripts are encoded using a hybrid dense-and-sparse model (BGE-M3 + BM25) to create robust textual indices. During retrieval, the assistant dually encodes a user’s query using both SigLIP and BGE-M3+BM25 text encoders. A concurrent search is then executed via the Hierarchical Navigable Small Worlds (HNSW) algorithm against all three vector databases (visual, caption, and audio). This tri-modal fusion allows our assistant to perform high-precision analysis and retrieval for complex, event-based queries.

Keywords: Multimedia Event Retrieval · Gemini · SigLIP · InternVL · ChunkFormer · TransNetV2 · HNSW · BGE-M3 · BM-25 · Ho Chi Minh City AI Challenge.

1 Introduction

The exponential growth of large-scale multimedia data on digital platforms has created an urgent need for efficient event retrieval systems. As the volume of video content expands, traditional search methods struggle to provide relevant results, as they often fail to comprehend the complex, multimodal nature of video. A user’s query may relate to visual semantics, specific spoken dialogue,

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or generated textual descriptions, creating a significant technical challenge. This problem has spurred numerous research challenges, from international competitions like the Lifelog Search Challenge (LSC) [11] and Video Browser Showdown (VBS) [12] to national-level initiatives like the Ho Chi Minh City AI Challenge [16].

To address this challenge, we propose GalaxyAssistant, an intelligent assistant framework designed for in-depth analysis and information retrieval from complex video data. Unlike monolithic systems, our system’s intelligence is rooted in its tri-modal indexing, which concurrently processes video via three parallel pipelines to build distinct, searchable databases for a video’s visual, captioned, and audio content.

Our methodology begins with preprocessing videos using TransNetV2 [1] for shot detection to extract representative keyframes. These keyframes and their corresponding audio tracks then feed into our three concurrent extraction pipelines: (1) A visual-language model (SigLIP [2]) generates dense semantic embeddings from keyframes for visual search. (2) An image captioning model (InternVL [3]) generates textual descriptions for the same keyframes. (3) An Automatic Speech Recognition (ASR) model, ChunkFormer, transcribes the video’s audio track. To create robust textual indices, both the generated captions and ASR transcripts are encoded using a hybrid dense-and-sparse model (BGE-M3 [4] + BM25 [5]).

During retrieval, the assistant dually encodes a user’s text query using both the SigLIP and BGE-M3+BM25 text encoders. A concurrent search is then executed via the Hierarchical Navigable Small Worlds (HNSW) [6] algorithm against all three vector databases (visual, caption, and audio). The primary contribution of this work is this tri-modal fusion approach, which allows our assistant to perform high-precision analysis and retrieval for complex, event-based queries by matching user intent against what is seen, described, and spoken.

The remainder of this paper is organized as follows. Section 2 discusses related work in multimodal retrieval. Section 3 details the architecture and implementation of GalaxyAssistant. Section 4 presents and explains how our assistant works. Section 5 outlines the limitations and future work. Finally, Section 6 concludes the paper.

2 Related Work

Content-based image and video retrieval (CBIR/CBVR) systems have become indispensable for navigating the exponential growth of multimedia data, finding critical applications in domains ranging from security surveillance and medical image analysis to large-scale academic benchmarks like the Video Browser Showdown (VBS) and Lifelog Search Challenge (LSC). Modern video retrieval has been revolutionized by large-scale Vision-Language Models (VLMs), with research primarily diverging into two main paradigms. The first is the joint-embedding approach, epitomized by CLIP [7] and its successors like ALIGN and BEiT-3. These models map text queries and video frames into a shared vector

space, enabling powerful cross-modal similarity search. While highly effective for matching objects and simple scenes, these models can struggle to capture complex temporal actions, subtle semantic relationships, or compositional logic. To address this, a second paradigm focuses on generation-based representation. Instead of joint-embedding, this approach "translates" visual content into rich text descriptions using advanced Captioning models like BLIP-2 [8] or specialized video-language models like Video-LLaMA [9]. These generated captions are then indexed for retrieval via text-to-text search. The critical weakness of this method, however, is its total dependency on the caption quality; any model "hallucination" or omission results in an irrecoverable retrieval failure. Furthermore, both vision-centric approaches often neglect the rich, parallel context provided by the Audio modality, which includes both speech (e.g., via ASR like Whisper) and non-speech audio events (e.g., via models like AudioSpectrogram Transformer). A significant research gap exists, as most systems commit to either a joint-embedding or a generative strategy and often treat audio as a mere auxiliary feature. We, therefore, propose a novel fusion architecture that synergizes the direct matching power of CLIP, the semantic richness of Captioning, and the essential context from Audio, achieving a robust balance of retrieval accuracy and deep semantic understanding.

3 Methodology

GalaxyAssistant is an advanced multimodal video retrieval system designed to address the challenges of large-scale event retrieval, motivated by competitions such as the Ho Chi Minh City AI Challenge. The system’s architecture has two primary phases: offline processing and online retrieval.

During the offline phase, videos undergo shot detection to extract representative keyframes. These keyframes, along with the original video audio, are then processed by three concurrent pipelines. The first pipeline uses a CLIP-based image encoder (ViT-SO400M-14-SigLIP-384) to generate dense visual embeddings, which are indexed in a Qdrant vector database. Concurrently, a second pipeline employs an image captioning model to produce textual descriptions; these are subsequently encoded and indexed using a hybrid dense-sparse approach (BGE-M3 + BM25). The third pipeline utilizes ChunkFormer to transcribe the audio, and the resulting text is indexed using the same hybrid BGE-M3 + BM25 method.

In the online retrieval phase, an user text query is first augmented using Gemini 2.5 Flash-Lite [13]. This enriched query is then separately encoded by the CLIP text encoder (for visual search) and the BGE-M3+BM25 encoders (for textual search). These query representations are used concurrently to search their respective collections. The visual search leverages the Hierarchical Navigable Small World (HNSW) algorithm within Qdrant to retrieve candidates based on CLIP similarity. For caption and audio modalities, Reciprocal Rank Fusion (RRF) is employed to fuse the results from their respective sparse (BM25) and dense (BGE-M3) retrievals. This retrieval process culminates in a multi-stage

fusion. First, the scores from the visual and caption results are normalized, combined, deduplicated, and sorted. This intermediate list is then re-ranked using the audio transcript (RRF) scores to produce the final output. This tri-modal fusion strategy, integrating visual semantics with both generated (caption) and transcribed (audio) textual information, enables high-precision retrieval of complex video events.

3.1 Data Processing

The initial data processing phase prepares the video corpus for indexing. Videos are first retrieved from the database and segmented into distinct scenes using TransNetV2. TransNetV2 is a deep neural network optimized for rapid and accurate shot transition detection, identifying cuts, fades, and dissolves that delineate scene boundaries. Leveraging this model provides an efficient and reliable method for partitioning the video content into meaningful segments suitable for subsequent analysis. From each detected scene, three keyframes are extracted corresponding to the beginning, middle, and ending points. This selection strategy offers a concise yet representative visual summary, capturing the scene’s entry, midpoint, and concluding state while balancing comprehensiveness and computational efficiency. These extracted keyframes, identified as representative snapshots of their respective scenes, are then stored in cloud object storage (AWS S3) for subsequent feature extraction.

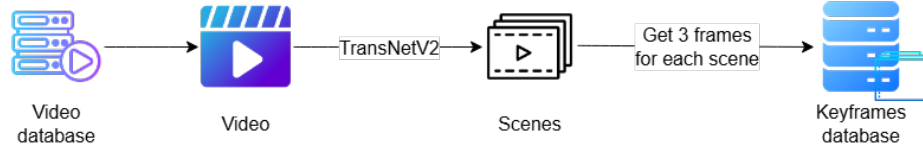


Fig. 1. Video Preprocessing phase

Following preprocessing, the vectorization phase transforms the extracted data into searchable embeddings, employing a hybrid approach that combines concept-based elements with embedding-based retrieval for enhanced efficiency. For visual representation, we utilize a CLIP-based model to enable text-based image retrieval. Specifically, we employ SigLIP, a state-of-the-art vision-language model. CLIP models, in general, learn a shared embedding space for images and text through contrastive training on large-scale image-text pairs. SigLIP improves upon the original CLIP by using a simpler sigmoid loss objective, which often leads to comparable or superior performance with increased training efficiency and scalability. Concurrently, the Vision-Language Model (VLM) InternVL is used to generate descriptive captions for the extracted keyframes, providing an alternative textual representation of the visual content.

Recognizing the importance of audio information, especially in news reports, documentaries, and educational videos prevalent in the dataset, we incorpo-

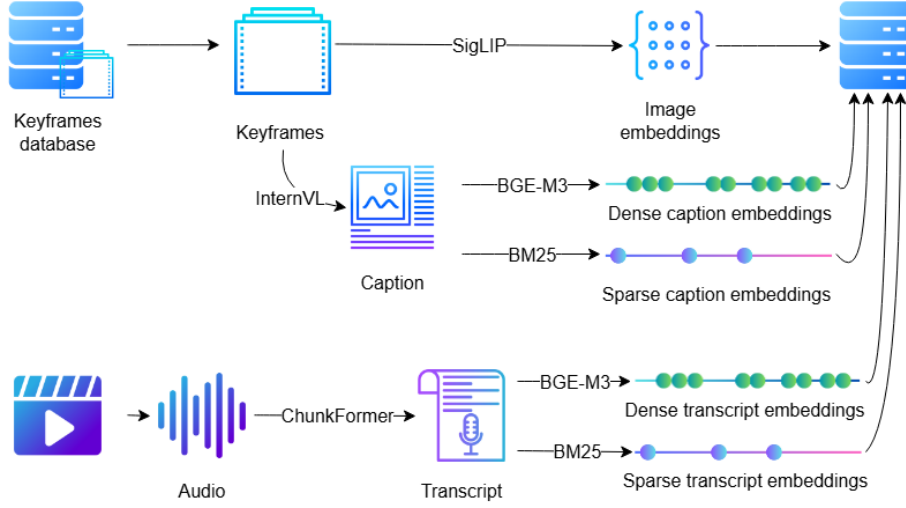


Fig. 2. Vectorization phase

rate Automatic Speech Recognition (ASR). We leverage ChunkFormer to transcribe spoken commentary and dialogue into text. ChunkFormer is a streaming Conformer-based model known for its efficiency in real-time ASR tasks. Compared to larger, more computationally intensive models, ChunkFormer offers significant advantages by being relatively lightweight while maintaining high effectiveness, particularly for languages like Vietnamese, balancing accuracy with lower latency and resource requirements.

Both the generated captions (from InternVL) and the audio transcripts (from ChunkFormer [10]) are then processed to create text embeddings using a hybrid strategy combining BGE-M3 and BM25. BGE-M3 is a powerful multilingual dense embedding model adept at capturing semantic meaning and context within the text. BM25, a classical sparse retrieval algorithm, complements this by focusing on keyword relevance and exact term matching. This dual encoding ensures that text search benefits from both semantic understanding (BGE-M3) and keyword precision (BM25).

Finally, all generated embeddings—visual embeddings from SigLIP, and both dense and sparse embeddings for captions and transcripts from BGE-M3 and BM25—are indexed into appropriate collections within Qdrant. Qdrant is a high-performance vector database optimized for efficient similarity search across large-scale, high-dimensional data. Its advantages include scalability, speed, support for hybrid dense-sparse search, and filtering capabilities, making it ideal for managing the diverse embeddings required by our system. This results in five distinct types of indexed embeddings: visual, dense caption, sparse caption, dense transcript, and sparse transcript, enabling comprehensive multimodal retrieval.

3.2 GalaxyAssistant

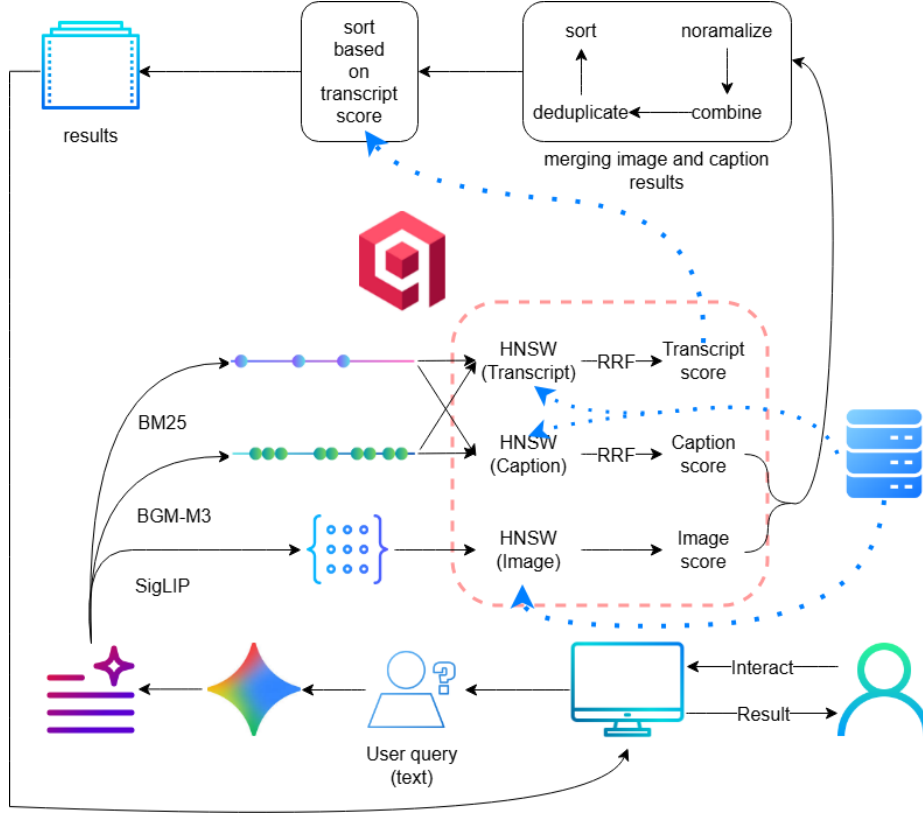


Fig. 3. Overview of GalaxyAssistant

The online retrieval phase of the GalaxyAssistant framework begins when a user submits a textual query via the interface. This query undergoes concurrent encoding through three distinct pathways: the SigLIP model generates a dense vector optimized for visual semantics, while both the BGE-M3 model (producing a dense semantic vector) and the BM25 algorithm (yielding a sparse keyword vector) process the text for textual matching. These query representations are then simultaneously utilized to search against the indexed vector databases within Qdrant. Specifically, the SigLIP vector queries the image embeddings, while the BGE-M3 and BM25 vectors perform hybrid searches against the caption and transcript embeddings, respectively. The underlying search mechanism leverages the Hierarchical Navigable Small Worlds (HNSW) algorithm for efficient retrieval. HNSW builds a multi-layered graph index that enables rapid traversal through the high-dimensional vector space. By utilizing long-range connections

in upper layers and short-range connections in lower layers, HNSW facilitates fast approximate nearest neighbor searches without exhaustive comparisons, significantly speeding up the retrieval process.

For caption and transcript retrieval, the hybrid search combines outputs from BGE-M3 and BM25. BGE-M3 excels at capturing semantic meaning but might miss specific keywords or rare terms crucial for relevance. Conversely, BM25 is strong at exact keyword matching but lacks understanding of synonyms or broader context. By fusing their results, the system leverages BGE-M3’s contextual understanding and BM25’s lexical precision, overcoming their individual limitations to achieve more robust and comprehensive text-based retrieval. The fusion of these dense and sparse search results is accomplished using Reciprocal Rank Fusion (RRF). RRF effectively combines the ranked lists by summing the reciprocal ranks of each item, prioritizing documents that rank highly in either the semantic or keyword search. This method provides a simple yet powerful way to merge the strengths of both retrieval strategies.

Following the concurrent searches, the results from the image and caption searches are merged, deduplicated, and their scores normalized. This combined list is then subjected to a final re-ranking step, where the scores derived from the transcript search are used to adjust the order, prioritizing results that exhibit strong relevance in the audio content. The finally ranked list of keyframes is then presented to the user, completing the retrieval process.

4 System Usage

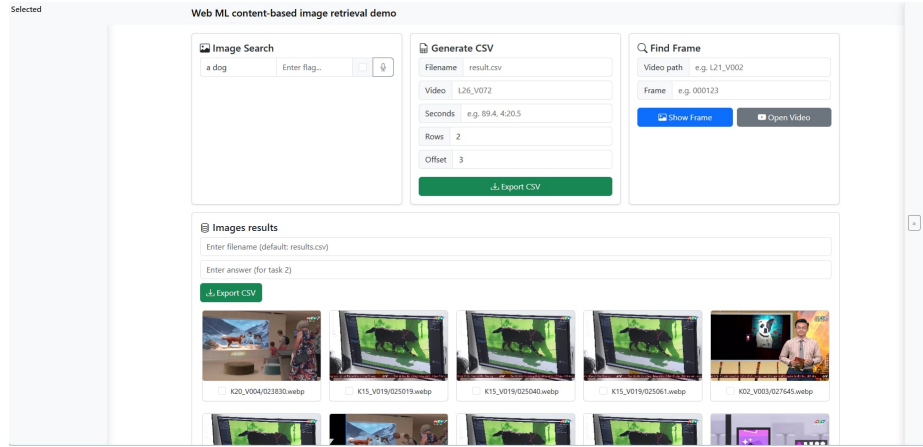


Fig. 4. Overview of GalaxyAssistant User Interface

While established systems for interactive video retrieval have demonstrated feature-rich paradigms [14], [15], such as temporal timeline browsing and

concept-based filtering, our approach prioritizes a minimalist, user-centered design. We posit that for multimedia retrieval problem, a simple and intuitive interface is paramount. GalaxyAssistant features a clean, web-based user interface (UI) designed for intuitive content-based image retrieval, reflecting a modern and minimalist approach. Users interact primarily through a straightforward "Image Search" panel to input text queries, with icons indicating support for voice commands, enhancing accessibility. Retrieval results are presented in the main "Images results" section as a grid of keyframe thumbnails, each labeled with its file path for easy identification. This central area also thoughtfully integrates tools for exporting results to CSV and providing answers for integrated Question Answering (Q&A) tasks. Complementing the core search, streamlined auxiliary panels offer utilities like "Find Frame" for direct access to specific video moments and "Generate CSV" for tailored data export, relevant for analysis or challenge submissions. Overall, the GalaxyAssistant UI provides a friendly and efficient environment, balancing powerful functionality with a clear, uncluttered design for effective query submission, result visualization, and task management.

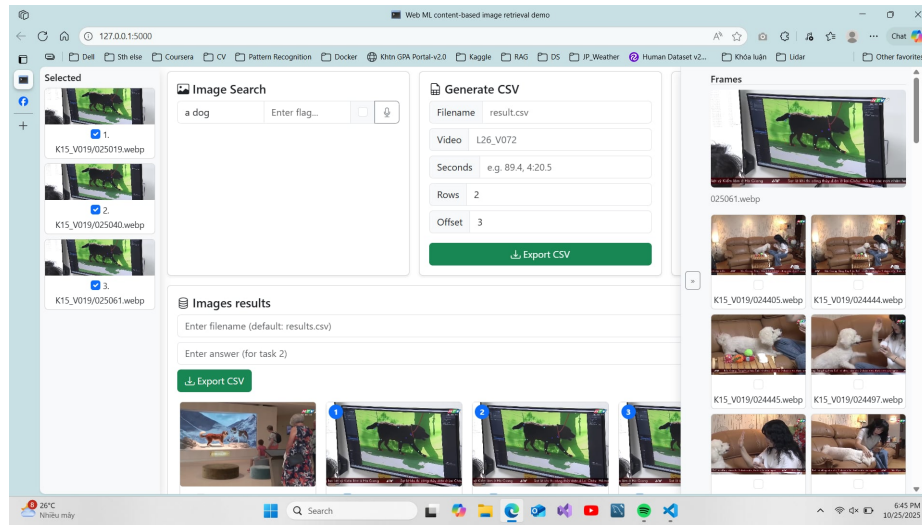


Fig. 5. User Interface when selecting the result

As users identify relevant keyframes within the main "Images results" grid, they can click to select them. Selected frames are visually marked and simultaneously compiled into a dedicated "Selected" sidebar on the left, providing a running list for review. Concurrently, interacting with a frame thumbnail dynamically updates a "Frames" sidebar on the right. This sidebar presents a larger preview of the selected frame, crucially accompanied by thumbnails of the 25 keyframes immediately preceding and the 25 keyframes immediately following it in the video sequence. This extended temporal context allows users to quickly

scan the surrounding moments, proving particularly useful for rapidly identifying and selecting multiple relevant frames clustered closely together in time.

5 Limitation

While the proposed GalaxyAssistant framework demonstrates significant potential for multimodal video retrieval, several limitations warrant consideration for future improvement. Firstly, the system’s overall efficacy is fundamentally dependent on the quality and scope of the underlying pre-trained models (SigLIP, InternVL, ChunkFormer, BGE-M3, BM25, Gemini). These models may exhibit inherent biases or encounter difficulties with out-of-domain content, specialized terminology, or visual and auditory nuances not adequately represented in their training data, particularly for lower-resource languages such as Vietnamese. Furthermore, inaccuracies introduced during preprocessing—including potential errors in shot detection (TransNetV2), suboptimal keyframe selection strategies that might miss brief but critical events, or imperfections in captioning and transcription—can propagate through the system and adversely affect final retrieval accuracy.

Secondly, the existing fusion and reranking logic, which primarily operates on score manipulation (normalization, RRF, transcript-based sorting), may not adequately capture the complex cross-modal relationships necessary for highly nuanced queries demanding simultaneous satisfaction of conditions across visual, textual, and auditory modalities. Thirdly, the framework currently does not integrate specialized models for Optical Character Recognition (OCR) or fine-grained object detection. This absence restricts its ability to leverage textual information embedded within frames or specific object attributes.

For instance, queries involving specific text (e.g., numbers on a pillar, as in the example provided: “Footage of many people standing around a stone pillar with the numbers 10, 12, 14, 16, 18, 20. Around this area, there are police officers. After a while, we see the cameraman standing on a bridge spanning a fast-flowing river.”) are less likely to be resolved accurately without OCR capabilities. Finally, the system currently lacks flexibility in combining different modality subsets (e.g., using only image and transcript, or image and caption). Implementing more adaptive fusion strategies could potentially yield more optimal results for queries where certain modalities are more relevant than others.

6 Conclusion

In conclusion, we presented GalaxyAssistant, an intelligent assistant framework for effective event retrieval from large-scale video collections. By employing a tri-modal indexing strategy that concurrently processes visual (SigLIP embeddings), generated textual caption (InternVL + BGE-M3/BM25 embeddings), and transcribed audio (ChunkFormer + BGE-M3/BM25 embeddings) information, our system creates a rich, searchable representation of video content. The retrieval phase leverages concurrent hybrid dense-and-sparse searches fused via

Reciprocal Rank Fusion (RRF) and executed against efficient vector indices, further enhanced by large language model-based query enrichment (Gemini) and an optional transcript-score-based reranking mechanism. This multifaceted approach allows GalaxyAssistant to effectively combine semantic understanding with lexical matching across modalities, demonstrating high potential for achieving precision in complex, event-based video queries, relevant for applications highlighted in challenges like the Ho Chi Minh City AI Challenge.

Future work should focus on addressing the identified limitations. For instance, refining the preprocessing pipeline, potentially through adaptive keyframe selection or domain-specific fine-tuning of the captioning and transcription models, could yield improvements in robustness. Furthermore, evaluating alternative query rewriting strategies and enhancing the user interface with advanced exploration tools also represent promising directions for future development.

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